

MPGNet

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Abstract

Methane, a greenhouse gas, has a 20-year Global Warming Potential (GWP) 80 times that of carbon dioxide, and has had its atmospheric concentration double over the past two centuries. While methane’s short atmospheric half-life (8–12 years) makes it an attractive target for mitigating global warming, monitoring its emissions presents significant challenges. Traditional ”bottom-up” surface monitoring methods are limited by access and cost, leading to underestimation of emissions. Satellite-based measurements offer global coverage but are hindered by temporal and spatial gaps caused by cloud cover and variable weather conditions. To address these challenges, generative models, particularly Conditional Generative Adversarial Networks (cGANs) and diffusion models, are gaining traction for their ability to simulate realistic methane plume behaviors from metadata inputs (e.g., emission rates, wind speed, and atmospheric pressure). These models can fill data gaps, generate synthetic plume maps, and provide stakeholders with visual insights to prioritize mitigation efforts. Previous applications of generative models in analogous domains, such as carbon dioxide dispersion and dissolved methane in aquifers, underscore their promise.

This project introduces a novel Methane Plume Generation Network (MPGNet) to synthesize methane plume maps using environmental variables, leveraging cGANs. As a work in progress, we hypothesize Denoising Diffusion Implicit Models to perform better.

1 Introduction

Methane emissions are a major contributor to global warming, with a global warming potential approximately 80 times higher than carbon dioxide over 20 years^[5]. Significant methane sources include petroleum and natural gas systems, landfills, and agriculture^[2]. With the introduction of methane mitigation programs, accurate reporting and monitoring have become critical^[8]. Traditional surface-level monitoring often underestimates emissions due to sparse measurements, while satellite-based methods face challenges such as cloud cover and insufficient temporal frequency^[4].

Generative models offer potential solutions for methane plume map generation by utilizing metadata such as wind speed, temperature, and emission rates^[7]. These models can generate synthetic plume dispersion maps that provide visualizations of methane concentra-

tion, enabling stakeholders to prioritize mitigation strategies effectively. Previous work has shown success in using conditional generative adversarial networks (cGANs) for predicting plume behaviors in other applications, including carbon dioxide dispersion^[5] and sub-surface flow modeling^[2].

In this project, we introduce the Methane Plume Generation Network (MPGNet) to generate methane plume dispersion maps using metadata and emission rates. Two generative approaches are considered: conditional generative adversarial networks (cGANs) and diffusion models. Due to time limitations and inference challenges associated with the deterministic DDIM sampling method, results for the diffusion model will not be reported here; instead, we focus on presenting its proposed architecture.

The ultimate goal is to leverage metadata and RGB map images to synthesize realis-

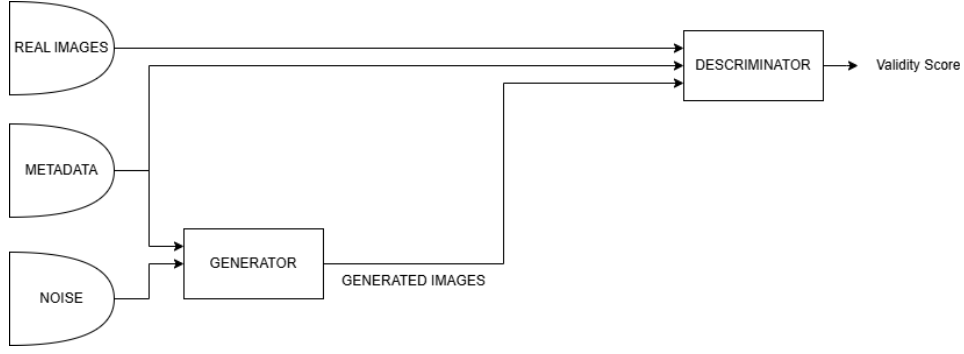


Figure 1: cGAN Architecture

tic plume maps, filling gaps in sparse data and enabling comprehensive visualization of

methane plume dispersion.

2 Model Architectures

2.1 cGAN

Conditional Generative Adversarial Network (cGAN) is comprised of two components: the **Generator** and the **Discriminator**. The generator is designed to synthesize high-quality plume map images conditioned on noise and metadata. We chose the cGAN specifically because its ability to condition image generation on metadata not only makes it ideal for controlling outputs based on specific plume characteristics but also provides a foundation for incorporating physics-based calculations in the future by leveraging the conditional features to model relationships between environmental variables and plume behaviors. Below is a detailed description of the architecture.

2.1.1 Generator

Dense Layer Transformation: Input noise and metadata are combined and passed through a linear (dense) layer. The output is normalized using **Batch Normalization** and activated using **ReLU** to ensure smooth gradients.

Transposed Convolution Layers: Feature maps are progressively upsampled using

transposed convolution layers. Each transposed convolution layer is followed by **Batch Normalization** and **ReLU** activation to maintain neuron diversity and produce non-negative activations.

ReLU was chosen over **LeakyReLU** to align with image generation tasks, ensuring non-negative pixel intensities and feature representations.

Tanh Activation: The final Convolutional block uses a **Tanh** activation which both normalizes and introduces non-linearity at the output layer, allowing the generator to model complex mappings from the latent space (metadata and noise). This enables the generator to model complex mappings from the latent space (metadata and noise).

2.1.2 Discriminator

Convolutional Layers: Progressive down-sampling layers extract hierarchical features from the input image. **LeakyReLU** activation is used to prevent "dead" neurons by allowing small gradients for negative inputs. This helps the discriminator learn subtle differences between real and generated images.

Metadata Integration: Metadata is processed through a fully connected layer, reshaped, and concatenated with the downsam-

pled image features. This enables the discriminator to evaluate image characteristics and their consistency with the metadata.

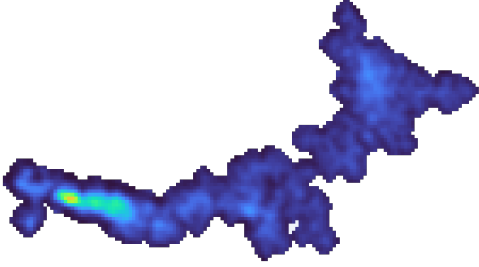


Figure 2: Plume Dispersion Map

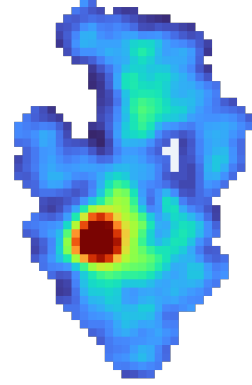


Figure 3: Plume Dispersion Map

3 Dataset

Images and tabular data used in this project originates from the Carbon Mapper Data Portal (2024), an open-source data platform that provides satellite images and measurements to quantify methane and carbon dioxide emissions. Methane and carbon dioxide column concentrations were gathered with satellite sensors, from which emission rates were quantified (Carbon Mapper, 2024). These were then used with satellite images to identify locations of plumes.

The dataset includes ortho-rectified whole scene images and plume dispersion maps (Figure 2). Images were captured under varying weather conditions, times of day, and geographic locations. Recorded environmental conditions include average wind speed and average wind direction, each with standard deviations. The IPCC emissions sector, date time, latitude and longitude of the plume are also included in the plume metadata. We selected a subset of approximately 5,076 plume map

images from the CarbonMapper^[1] dataset, along with associated metadata stored in multiple CSV files within the Data/Metadata/ directory. For pre-processing, we aggregated metadata from all CSV files and handled missing values by imputing numerical features with their respective means. Wind direction angles were transformed using sine and cosine functions to capture their cyclical nature. Plume boundary data were standardized to ensure consistency, and all images were resized to 64×64 pixels and normalized to a $[-1, 1]$ range to facilitate stable training. Additionally, image transformations such as resizing and normalization acted as data augmentation techniques, enhancing the diversity of training samples. That being said, the dataset may exhibit potential biases, such as over-representation of certain plume conditions or geographical areas, which did limit the model’s ability to generalize to underrepresented scenarios. Furthermore, there was often missing or inaccurately labeled metadata, which introduced unwanted noise, affecting the quality of the generated plume maps.

4 Training

For the training process we used PyTorch (with *CUDA* hardware acceleration), leveraging its frameworks for neural network construction and optimization. The optimization was performed using the Adam optimizer (for both models).

A learning rate of 0.0002 and hyper-parameters β_1 and β_2 were set to 0.5 and 0.999, respectively, which are standard choices for stabilizing GAN training. Key hyper-parameters included a batch size of 64, a noise dimension of 100, and a feature map size of 64. The noise vector serves as a source of randomness that allows the generator to produce a wide variety of outputs even when conditioned on the same metadata. Without this noise, the generator would produce a single deterministic output for each set of metadata, severely limiting the diversity and richness of the generated plume maps. To prevent over-fitting, we incorporated normalization layers within the generator and discriminator architectures and maintained a consistent data augmentation strategy through image transformations such as resizing and nor-

malization. Although explicit regularization techniques like dropout were not directly applied, the use of a balanced dataset and careful hyper-parameter tuning contributed to model generalization. Throughout training, we monitored the discriminator and generator loss values, printing their averages at each epoch to track convergence and performance.

Something that we observed during the training process was that the discriminator would often become too effective at distinguishing real plumes from the generated (fake) ones. One approach we tried to prevent this over fitting was to add a small amount of Gaussian noise to the 'real' images. This was done in order to make discriminator's job harder, forcing the generator to improve. Despite a decrease in generator loss and a slight increase in discriminator loss, artifacts in the generated images led us to revert to our previous approach. We also added dropout to the discriminator which randomly deactivated a subset of neurons during training, preventing the discriminator from over fitting.

5 Performance and Evaluation

5.1 cGAN

The performance of the cGAN model was evaluated using both quantitative and qualitative metrics. Quantitatively, we used adversarial loss trends for the generator and discriminator to monitor training stability and Fréchet Inception Distance (FID) to assess the similarity between generated plume maps and real images. While comparisons to state-of-the-art cGAN models were limited due to the domain-specific nature of our dataset, the generated maps demonstrated competitive quality based on visual inspection. Qualitatively, visual examples of generated plume maps were compared to real counterparts, highlighting

the model's ability to replicate realistic emission patterns and variations. However, limitations such as occasional artifacts in generated maps and reduced diversity in under-represented conditions were noted. Future improvements include fine-tuning the generator's architecture, exploring more robust loss functions, and incorporating additional diverse data to enhance generalization and quality.

5.2 DDIM

As a work in progress, the DDIM^[6] diffusion model would be used to make higher quality plumes using the proposed architecture below.

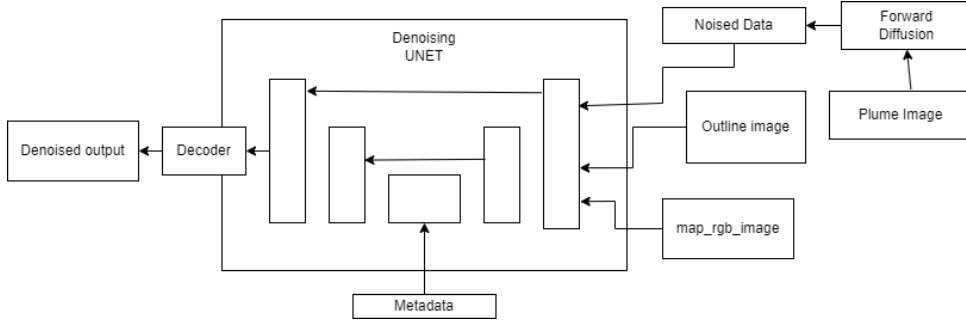


Figure 4: DDIM UNet

6 Discussion

Unlike prior work, such as the study done on [hyper spectral remote sensing](#)^[3], our approach is uniquely tailored to generating plume maps for visualization purposes. This makes our method particularly valuable for stakeholders and investors, as it provides an intuitive and visually interpretable way to understand complex environmental scenarios and emissions data. Future research could build on this by incorporating interactive visualizations or coupling with predictive models to further enhance decision-making processes.

One limitation we encountered was that the metadata was not as useful as we initially expected. For example, there was no information about the exact coordinates of the plant’s stack, meaning that we could not accurately determine where the plume originated. Furthermore, while positional coordinates were available, they did not account for the rotational component of plume dispersion. As a result, the generated plumes often exhibited random orientations, even though the plume size was accurate.

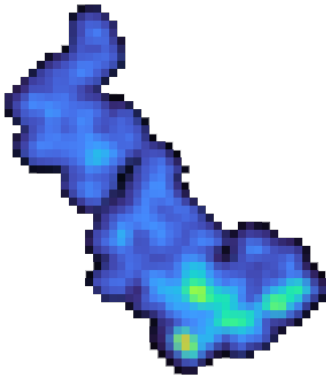


Figure 5: Original Plume



Figure 6: cGAN Generated Plume

7 Conclusion

We conclude that generation models are really powerful in looking at a small data sample (almost 4000 samples) and creating plume maps from just the environmental data. This approach

offers a cost-effective solution for generating plume maps, providing valuable insights to support stakeholders and investors in visualizing environmental impacts. By leveraging these models, decision-makers can better identify and evaluate carbon mitigation strategies, ultimately facilitating informed planning and sustainable investment.

One consideration is that the dataset size may be insufficient for effectively training the DDIM^[6] model, potentially limiting its performance.

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